



Modeling of Single-Phase Flows in Porous Media and Passive Contaminant Transport

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Presentation structure

- ➊ General definitions
- ➋ Single-phase flow modeling
- ➌ Finite volume method for elliptic equations
- ➍ Introduction to passive transport problems in porous media
- ➎ Finite volume method for hyperbolic equations
- ➏ References

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What is a porous medium?

A porous medium is characterized by its **porosity** ϕ , the ratio between the volume of void spaces V_p and the total volume V_t :

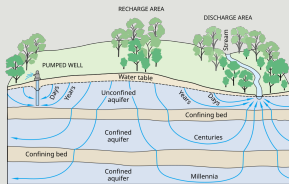
$$\phi = \frac{V_p}{V_t},$$

thus being a dimensionless quantity between zero and one.



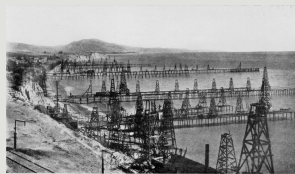
(a) Limestone (Torcal de Antequera)

Source: Fernando Domínguez Cerejido /
Wikimedia Commons (CC BY-SA 4.0)



(b) Underground aquifer

Source: T.C. Winter et al., U.S. Geological
Survey (Public Domain)



(c) Oil reservoir

Source: G.H. Eldridge (c. 1905) / NOAA
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Figure: Examples of porous media.

Compressibility and viscosity

Media can be considered:

- **Incompressible**, if they have *static* porosity; and
- **Compressible**, if they have *dynamic* porosity.

Flows, in turn, can be classified as:

- **Incompressible**: has zero compressibility, implying a constant density independent of pressure;
- **Slightly compressible**: has a low and constant compressibility, with its density varying linearly with pressure; or
- **Compressible**: has high compressibility and its density increases with pressure, tending to stabilize at high pressures.

Viscosity is the *measure of a fluid's resistance to its own flow*. Gases exhibit low viscosity, and denser fluids (oil) exhibit high viscosity.

Permeability of the medium

Permeability K represents the *ability* of the porous medium to *transmit a fluid through its interconnected pores*, when fully saturated with it. When only a single fluid is present in the porous medium, this property is called **absolute permeability**.

Permeability is given by a *symmetric positive definite* tensor:

$$K = \begin{bmatrix} K_{xx} & K_{xy} & K_{xz} \\ K_{yx} & K_{yy} & K_{yz} \\ K_{zx} & K_{zy} & K_{zz} \end{bmatrix},$$

if it is **isotropic**, K is a scalar.

When K is spatially constant, the medium is called **homogeneous**; otherwise, it is **heterogeneous**.

Darcy's Law

To introduce fluid flow modeling, one must first understand the **Darcy velocity**. This equation stems from studies of the physical flow of water in a vertical sand filter conducted by Henry Darcy in 1856.

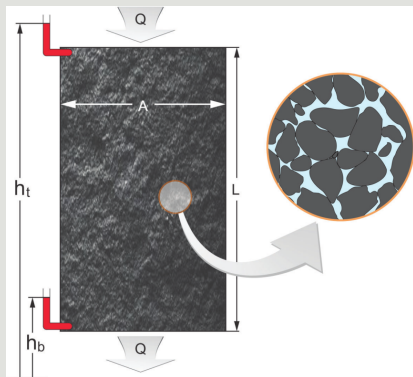


Figure: Darcy's experiment, SOUZA and ROCHA (2022, p. 16).

The flow rate Q (volume per time) through a cross-sectional area A defines the **Darcy velocity**:

$$u = \frac{Q}{A} = -\frac{K}{\mu}(\nabla p - \rho g \nabla z), \quad (1.1)$$

where z is depth. This equation (1.1) is known as **Darcy's Law** and represents the conservation of momentum, with pressure and gravity driving the flow.

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Single-phase flow modeling

When pressure (or flow) is applied to a domain saturated by only **one fluid**, what is called **single-phase flow** is induced. The conservation of mass of the fluid implies a balance law:

$$\frac{\partial}{\partial t} \int_V \phi \rho \, dx + \int_{\partial V} \rho (u \cdot n) \, ds = \int_V q \, dx. \quad (2.1)$$

Applying the divergence theorem, using Darcy's Law and developing it, we obtain the partial differential equation with unknowns p and ρ :

$$\rho \phi c_t \frac{\partial p}{\partial t} - \nabla \cdot \left(\frac{\rho K}{\mu} (\nabla p - \rho g \nabla z) \right) = q, \quad (2.2)$$

where $c_t = c_f + c_r$ represents the total compressibility. Considering an **incompressible flow**, equation (2.2) becomes an elliptic equation with variable coefficients:

$$- \nabla \cdot \left(\frac{\rho K}{\mu} (\nabla p - \rho g \nabla z) \right) = q. \quad (2.3)$$

Auxiliary conditions

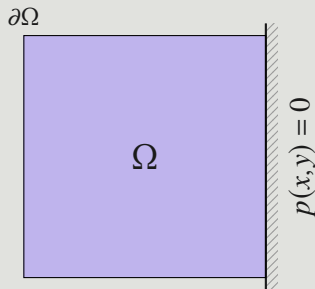
In addition to the equations that actually model single-phase flows, boundary conditions are also necessary. These conditions can be mainly classified into three types:

- **Dirichlet**, where pressure $p(x)$ is given;
- **Neumann**, where the flow $\nabla u \cdot n$ is given; and
- **Robin** or mixed, where $\alpha u + \beta(\nabla u \cdot n)$ is specified;

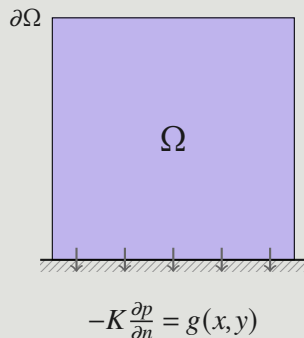
where u is a scalar function, such as temperature or pressure. These conditions can be **homogeneous**, when the boundary is equal to zero, or **heterogeneous**, when they are equal to a function g .

Auxiliary conditions

To exemplify the types of boundaries on the edges $\partial\Omega = \partial\Omega_p \cup \partial\Omega_u$, the following figures are presented:



(a) Homogeneous Dirichlet boundary.



(b) Non-homogeneous Neumann boundary.

Figure: Exemplification of boundary types.

Homogeneous and heterogeneous media

To classify media as homogeneous or heterogeneous, a simplification of (2.3) for single-phase flow is considered, with the hypotheses that the flow is *incompressible, isothermal, and without gravitational effect*, such that:

$$\left\{ \begin{array}{ll} \nabla \cdot u = \frac{q}{\rho} & \text{in } \Omega \\ u = -\frac{K}{\mu} \nabla p & \text{in } \Omega \\ p = p_b & \text{on } \partial\Omega_p \\ u \cdot n = u_b & \text{on } \partial\Omega_u \end{array} \right., \quad (2.4)$$

where viscosity μ and density ρ are uniform in the incompressible case and considered unitary here. Also, where $\partial\Omega = \partial\Omega_p \cup \partial\Omega_u$ is the boundary of the domain $\Omega \subset \mathbb{R}^d$ with $d = \{1, 2, 3\}$.

Homogeneous and heterogeneous media

Therefore, the medium can be classified mainly between:

- A so-called **homogeneous** medium, governed by the single-phase flow system (2.4), occurs when the absolute permeability K is constant and uniform, thus not depending on x ; or
- When the medium is **heterogeneous**, the absolute permeability K varies with x and then the flow tends to pass through regions of high permeability and avoid those of low permeability.

Also, it is possible to use a model from the *SPE10* project provided by the *Society of Petroleum Engineers*, used as a reference in oil reservoir simulations.

Finite volume method

The **finite volume method** is a discretization method useful for numerical simulations of conservation laws of various types: *elliptic*, *hyperbolic*, or *parabolic*, for example. This method has some important points:

- It can be used in **arbitrary geometries**;
- It can be used on **structured or unstructured grids**; and
- It **locally conserves numerical fluxes**, which makes it particularly interesting for fluid mechanics problems.

This can be achieved because it is based on a "balanced" approach:

- A local balance is written in each discretization cell, which is often called a *control volume*;
- By the divergence formula, an integral formulation of the fluxes over the boundary of the control volume is obtained.

The fluxes at the boundaries are discretized with respect to the discrete "unknowns".

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Finite volume method for elliptic equations

The objective of this section is to numerically approximate the solution of system (2.3) with simplifications:

$$\left\{ \begin{array}{ll} -\nabla \cdot (K \nabla p) = q & \text{in } \Omega \\ p = p_b & \text{on } \partial\Omega_p, \\ (-K \nabla p) \cdot n = u_b & \text{on } \partial\Omega_u \end{array} \right. \quad (3.1)$$

in the case of single-phase flow, with the hypotheses of being incompressible, isothermal, and without gravitational effect, with constant and unitary viscosity μ and density ρ .

Moreover, the pressure is related to the Darcy velocity by

$$u = -K \nabla p. \quad (3.2)$$

One-dimensional case

Considering a one-dimensional domain $\Omega = [a, b]$ where the first equation of (3.1) is written as

$$\frac{d}{dx} \left(K \frac{dp}{dx} \right) = q. \quad (3.3)$$

With finite volume discretization, we can elaborate in each volume V_i :

$$-\frac{1}{\Delta x^2} \left(K_{i+1/2}(p_{i+1} - p_i) - K_{i-1/2}(p_i - p_{i-1}) \right) = q_i. \quad (3.4)$$

To implement the finite volume method, we write (3.4) in matrix form

$$Aw = d, \quad (3.5)$$

where w and d represent the vectors of approximate values for pressure and injection q , respectively. Finally, A is a tridiagonal matrix defined by

$$(Aw)_i = \frac{1}{\Delta x^2} \left(-K_{i-\frac{1}{2}}p_{i-1} + \left(K_{i-\frac{1}{2}} + K_{i+\frac{1}{2}} \right) p_i - K_{i+\frac{1}{2}}p_{i+1} \right), \quad (3.6)$$

for $i = 2, \dots, N-1$. In the context of a cell-centered grid, the rows $i = 1$ and $i = N$ are used to impose boundary conditions.

One-dimensional case: Implementation

A **Neumann boundary** can be imposed when $i = 1$, so that the left boundary condition would be at cell $x_{i-1/2} = x_{1/2}$. Then, with

$$-K \left. \frac{dp}{dx} \right|_{x_{\frac{1}{2}}} = u_b,$$

for the first equation of system (3.5):

$$\frac{1}{\Delta x^2} \left(-K_{\frac{3}{2}} (p_2 - p_1) \right) = q_1 + \frac{u_b}{\Delta x}.$$

On the other hand, when $i = N$, at face $x_{N+1/2}$, a **Dirichlet boundary** with imposed pressure can be used

$$p|_{x_{N+1/2}} = p_b.$$

Therefore, using a discretization with half a control volume, the last equation of system (3.5) becomes:

$$\frac{1}{\Delta x^2} \left(2K_N p_N + K_{N-\frac{1}{2}} (p_N - p_{N-1}) \right) = q_1 + \frac{2K_N p_b}{\Delta x^2}.$$

One-dimensional case: Continuous example

Example

Given a one-dimensional boundary value problem

$$\begin{cases} -\frac{d}{dx} \left(K \frac{dp}{dx} \right) = -25 \cos(25x) & \text{in } \Omega = [0, 1] \\ p = x & \text{on } \partial\Omega \end{cases},$$

where the absolute permeability of the medium is $K(x) = 2 + \sin(25x)$ and exact solution $p(x) = x$. The factorizations produced results close to the exact solution, with truncation error $\epsilon \approx 0.355 \times 10^{-2}$, and plots:

One-dimensional case: Continuous example

Example

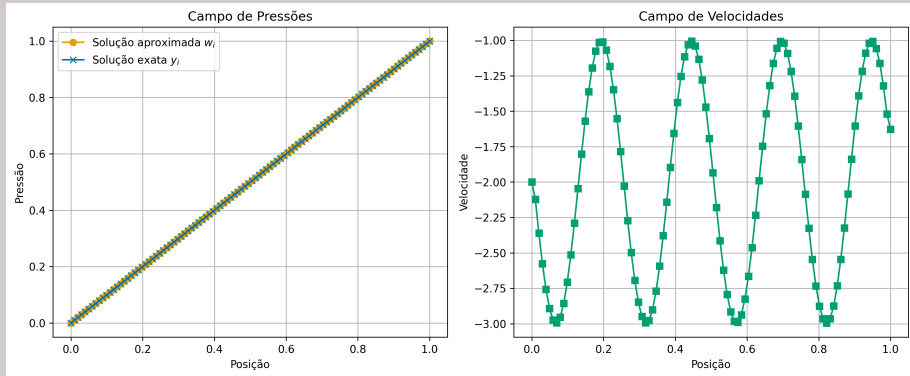


Figure: Pressure and velocity fields of the example.

One-dimensional case: Analysis of factorization methods

We performed tests with the three factorization algorithms: **LDL^T**, **Crout** and **Cholesky**. We see that the **LDL^T** method performed better than Crout factorization (considering the problem of the example in this section).

Subintervals	LDL ^t	Crout	Cholesky
10^2	$0.18247 \times 10^{-4} s$	$0.18758 \times 10^{-4} s$	$0.20147 \times 10^{-4} s$
10^3	$0.86234 \times 10^{-4} s$	$0.95469 \times 10^{-4} s$	$0.98242 \times 10^{-4} s$
10^4	$0.82834 \times 10^{-3} s$	$0.92428 \times 10^{-3} s$	$0.94293 \times 10^{-3} s$
10^5	$0.72082 \times 10^{-2} s$	$0.79367 \times 10^{-2} s$	$0.83214 \times 10^{-2} s$

Table: Average times of different factorization methods.

Two-dimensional case

The generalization of the finite volume method to a higher dimension follows an analogous procedure. At the end of the development, the discrete form for a two-dimensional problem is

$$\begin{aligned} q_{ij} = & -\frac{1}{\Delta y^2} K_{i,j+\frac{1}{2}} p_{i,j+1} - \frac{1}{\Delta y^2} K_{i,j-\frac{1}{2}} p_{i,j-1} \\ & - \frac{1}{\Delta x^2} K_{i+\frac{1}{2},j} p_{i+1,j} - \frac{1}{\Delta x^2} K_{i-\frac{1}{2},j} p_{i-1,j} \\ & + \left(\frac{1}{\Delta y^2} K_{i,j+\frac{1}{2}} + \frac{1}{\Delta y^2} K_{i,j-\frac{1}{2}} + \frac{1}{\Delta x^2} K_{i+\frac{1}{2},j} + \frac{1}{\Delta x^2} K_{i-\frac{1}{2},j} \right) p_{ij}, \end{aligned} \quad (3.7)$$

where harmonic means are used for $K_{i,j\pm 1/2}$ and $K_{i\pm 1/2,j}$.

After the pressure is calculated, the velocity field can be computed using the same strategy of approximating the fluxes in the integrals over ∂V_{ij} :

$$\begin{aligned} u_L & \simeq -K_{i+\frac{1}{2},j} \frac{p_{i+1,j} - p_{ij}}{\Delta x}, & u_O & \simeq -K_{i-\frac{1}{2},j} \frac{p_{ij} - p_{i-1,j}}{\Delta x} \\ v_N & \simeq -K_{i,j+\frac{1}{2}} \frac{p_{i,j+1} - p_{ij}}{\Delta y}, & v_S & \simeq -K_{i,j-\frac{1}{2}} \frac{p_{ij} - p_{i,j-1}}{\Delta y}. \end{aligned}$$

Two-dimensional case: Implementation

Considering an $M \times N$ computational cell discretization, the scheme (3.7) can be written in matrix form

$$Aw = d$$

with $w = (p_1, \dots, p_{MN})^t$ and $b = (q_1, \dots, q_{MN})^t$. Each row of matrix A is related to a cell (i, j) and takes into account the contributions of its four neighbors. For example, in a 3×3 grid, the cells can be ordered *by rows* from left to right, top to bottom:

$i-1, j-1$ 7	i, j 8	$i+1, j+1$ 9
$i-1, j-1$ 4	i, j 5	$i+1, j+1$ 6
$i-1, j-1$ 1	i, j 2	$i+1, j+1$ 3

Figure: Example of ordering for a two-dimensional problem.

Two-dimensional case: Implementation

In this way, with this ordering, the matrix will have a pentadiagonal structure, three successive diagonals and two at a distance M from the main diagonal:

$$A_{9 \times 9} = \begin{bmatrix} \times & \times & & \times & & & & & \\ \times & \times & \times & & \times & & & & \\ & \times & \times & & & \times & & & \\ \times & & & \times & \times & & \times & & \\ & \times & & \times & \times & \times & & \times & \\ & & \times & & \times & \times & & & \times \\ & & & \times & & \times & \times & & \\ & & & & \times & & \times & \times & \times \\ & & & & & \times & \times & \times & \times \end{bmatrix}.$$

This ordering can be calculated by an algebraic relation given by

$$k = i + (j - 1)M,$$

where k is the unknown number and corresponding matrix row, M the number of cells in each row of the grid, $i = 1, 2, \dots, M$ and $j = 1, 2, \dots, M$.

Two-dimensional case: Example

Definition

In a reservoir, when a portion of this place is sectioned into a domain Ω with exactly one injection well and one production well (where a fluid is extracted), this portion is called **a quarter of the five spot**. Injection and production are induced by a source term q

$$q = \begin{cases} \tilde{q} & \text{at the injection well} \\ -\tilde{q} & \text{at the production well} \\ 0 & \text{in the rest of the domain} \end{cases},$$

applied in cells located at extremes of Ω , usually injection at the bottom left extreme and production at the top right.

Two-dimensional case: Example

Example

Given a simplified two-dimensional elliptic problem as (3.1), a **a quarter of the five spot** configuration as in definition 3 and homogeneous Neumann boundary conditions:

$$\left\{ \begin{array}{ll} \nabla \cdot u = q & \text{in } \Omega = [0, 1] \times [0, 1] \\ u \cdot n = 0 & \text{on } \partial\Omega \\ -K\nabla p = u & \text{(Darcy velocity)} \end{array} \right. .$$

With a constant absolute permeability $K(x) = 1$, source term $\tilde{q} = 1$, injection well at cell $(1, 1)$ and production well at (N, M) .

To solve the problem accurately, a resource of **local Dirichlet boundary** was used, where a pressure is imposed at a specific point of the matrix, in this case, point $(1, 1)$ for simplicity.

Two-dimensional case: Implementation

Example

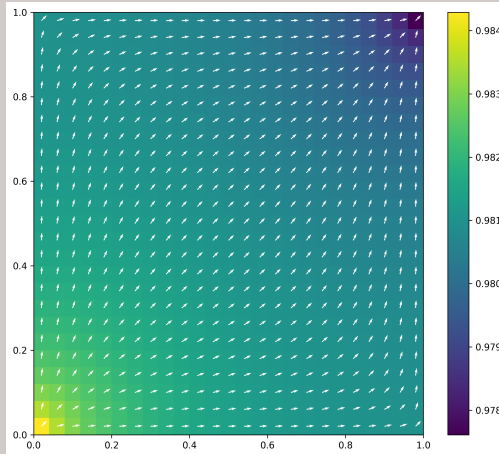


Figure: Pressure field and vectors of the example.

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Passive transport problems in porous media

The movement of a fluid in single-phase flow through porous media is described by a *passive transport problem*, where the **marked fluid** follows the flow *without altering its properties*.

With the fluid velocity u obtained from the solution of the elliptic problem given by (3.1) and (1.1), this displacement can be estimated by the following hyperbolic equation:

$$\left\{ \begin{array}{ll} \frac{\partial}{\partial t}(\phi c) + \nabla \cdot (uc) = q & \text{in } \Omega \\ c(x, t = 0) = c_0 & \text{in } \Omega \\ c(x, t) = c_b(x, t) & \text{on } \partial\Omega^- \end{array} \right. , \quad (4.1)$$

where ϕ is porosity, q is the source term, $c(x, t)$ is the contaminant concentration, c_0 is the initial condition, and c_b is the concentration at the inflow boundaries

$$\partial\Omega^- = \{x \in \partial\Omega; u \cdot n < 0\},$$

where n is the outward unit normal to the boundary $\partial\Omega$.

Passive transport problems in porous media

In numerical simulations, the source term in equation (4.1) usually accounts for injection and production wells, which can be converted into appropriate boundary conditions, leading to the equation:

$$\begin{cases} \phi \frac{\partial c}{\partial t} + \nabla \cdot (uc) = 0 & \text{in } \Omega \\ c(x, t = 0) = c_0 & \text{in } \Omega \\ c(x, t) = c_b(x, t) & \text{on } \partial\Omega^- \end{cases}, \quad (4.2)$$

where porosity $\phi = \phi(x)$ is constant in time. If porosity is uniform and constant in time, the first equation in (4.2) can be scaled to

$$\frac{\partial c}{\partial \tau} + \nabla \cdot (uc) = 0, \quad (4.3)$$

where $\tau = t/\phi$. Assuming that the velocity u is known and does not depend on concentration (considering K constant), we have a **linear hyperbolic conservation law**.

Derivation of hyperbolic conservation laws

Here, a **balance law** is derived to determine the conservation of *concentration* $c(x, t)$ in a domain $\Omega \subset \mathbb{R}^n$. This balance law states that the temporal variation of the quantity c in a domain Ω equals the flux rate of c through $\partial\Omega$ plus the total c injected into or extracted from Ω :

$$\frac{\partial}{\partial t} \int_{\Omega} \phi c(x, t) \, dx = - \int_{\partial\Omega} f(c(x, t)) \cdot n \, ds + \int_{\Omega} q \, dx, \quad (4.4)$$

where n is the outward unit normal to $\partial\Omega$, $f(c)$ is the flux function depending on c (not necessarily linear), and q is the source term. Using the divergence theorem:

$$\int_{\Omega} \left(\frac{\partial}{\partial t} (\phi c(x, t)) + \nabla \cdot f(c(x, t)) - q \right) dx = 0. \quad (4.5)$$

Since equation (4.5) holds for any arbitrary domain Ω and porosity $\phi = \phi(x)$ is constant in time, the differential form can be obtained:

$$\phi \frac{\partial c}{\partial t} + \nabla \cdot f(c) = q \quad (4.6)$$

which is then called a **hyperbolic partial differential equation**.

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Finite volume method for hyperbolic equations

Considering a hyperbolic conservation law of the form (4.6) with $\phi = 1$ and $q = 0$, we have the one-dimensional form

$$\frac{\partial c}{\partial t} + \frac{\partial}{\partial x} f(c) = 0. \quad (5.1)$$

At each t^n , we define:

$$C_i^n = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} c(x, t^n) dx \quad \text{and} \quad \bar{F}_{i\pm\frac{1}{2}}^n = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(c(x_{i\pm\frac{1}{2}}, t)) dt.$$

Integrating the conservation law (5.1) over $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t^n, t^{n+1})$, separating the integrals and using the fundamental theorem of calculus, we obtain:

$$C_i^{n+1} = C_i^n - \frac{\Delta t}{\Delta x} \left(\bar{F}_{i+\frac{1}{2}}^n - \bar{F}_{i-\frac{1}{2}}^n \right). \quad (5.2)$$

That is, the equation above establishes a conservation principle: *the average variation of concentration in the cell is given by the difference of the fluxes at its boundaries*; still without any approximations.

Upwind method for approximating discrete fluxes

To approximate the discrete fluxes $F_{i\pm 1/2}^n$, various methods can be used, such as:

- *Central scheme* (finite differences);
- *Lax-Friedrichs* method; or
- *Lax-Wendroff* method.

The method used in this work will be the **upwind method**, which takes into account the *structure of the solution*, so that the information at each point is obtained by looking in the direction from which it propagates.

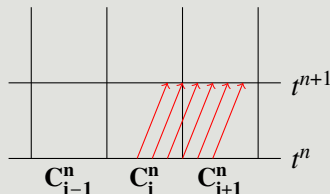


Figure: Propagation of information in a cell.

Upwind method: One-dimensional case

Considering $u^+ = \max\{u, 0\}$ and $u^- = \min\{u, 0\}$, it is possible to discretize (5.2) in the case of **constant velocities**:

$$C_i^{n+1} = C_i^n - \frac{\Delta t}{\Delta x} (u^+ C_i^n + u^- C_{i+1}^n - u^+ C_{i-1}^n - u^- C_i^n). \quad (5.3)$$

Now, in the case of **variable velocities**:

$$\begin{aligned} C_i^{n+1} = C_i^n - \frac{\Delta t}{\Delta x} & \left[\left(u^+(x_{i+\frac{1}{2}}) - u^+(x_{i-\frac{1}{2}}) + u^-(x_{i+\frac{1}{2}}) - u^-(x_{i-\frac{1}{2}}) \right) C_i^n \right] \\ & - \frac{\Delta t}{\Delta x} \left[u^-(x_{i-\frac{1}{2}}) W_{i-\frac{1}{2}} + u^+(x_{i+\frac{1}{2}}) W_{i+\frac{1}{2}} \right]. \end{aligned} \quad (5.4)$$

Also, it is necessary to satisfy the **CFL condition**, a necessary but not sufficient condition to guarantee convergence of the finite volume method for the differential equation. Convergence occurs if:

$$u \frac{\Delta t}{\Delta x} \leq 1,$$

with u constant.

Upwind method: Example of the one-dimensional case

Example

Given the one-dimensional boundary value problem (3.1), which models single-phase flow in a saturated reservoir, with the system of equations given by:

$$\begin{cases} -\frac{d}{dx} \left(K \frac{dp}{dx} \right) = -25 \cos(25x) & \text{in } \Omega_e = [0, 1] \\ p = x & \text{on } \partial\Omega_e \end{cases} \quad (5.5)$$

where $K(x) = 2 + \sin(25x)$. The flow of a contaminant with concentration $c(x, t)$ will be simulated, governed by the system:

$$\begin{cases} \partial_t c + u \partial_x c = 0 & \text{in } \Omega_h \\ c_0(x) = e^{-20(3x-.5)^2} + e^{-(3x-3.5)^2} & \text{in } \Omega_h \end{cases},$$

Using equation (5.4), it is possible to arrive at a solution visualized by the following plots:

Upwind method: Example of the one-dimensional case

Example

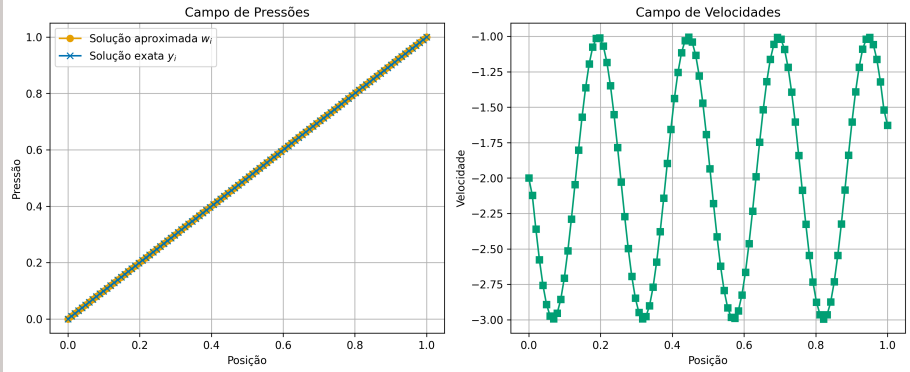


Figure: Pressure and velocity fields of the example.

Upwind method: Example of the one-dimensional case

Example

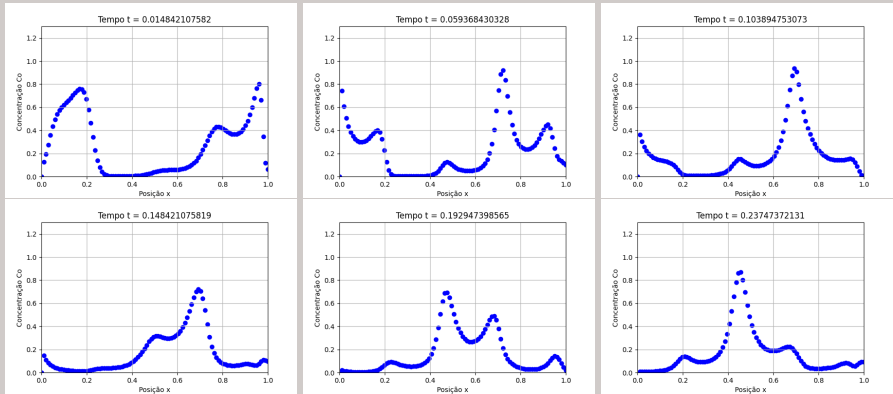
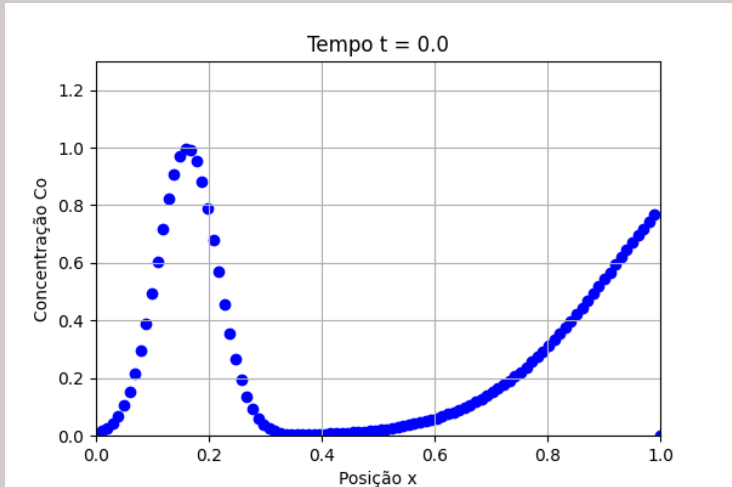


Figure: Concentration plots at different times.

Upwind method: Example of the one-dimensional case

Example



Upwind method: Two-dimensional case

For two-dimensional problems, the conservation law (5.1) takes the form

$$c_t + f_x(c) + g_y(c) = 0, \quad (5.6)$$

where the fluid concentration depends on x , y , and t . Defining

$$\bar{F}_{i+\frac{1}{2},j}^n = \frac{1}{\Delta t \Delta y} \int_{t^n}^{t^{n+1}} \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} f(c(x_{i+\frac{1}{2}}, y, t)) dy dt \quad (5.7)$$

and

$$\bar{G}_{i,j+\frac{1}{2}}^n = \frac{1}{\Delta t \Delta x} \int_{t^n}^{t^{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} g(c(x, y_{j+\frac{1}{2}}, t)) dx dt, \quad (5.8)$$

we have

$$C_{i,j}^{n+1} = C_{i,j}^n - \frac{\Delta t}{\Delta x} \left(\bar{F}_{i+\frac{1}{2},j}^n - \bar{F}_{i-\frac{1}{2},j}^n \right) - \frac{\Delta t}{\Delta y} \left(\bar{G}_{i,j+\frac{1}{2}}^n - \bar{G}_{i,j-\frac{1}{2}}^n \right) \quad (5.9)$$

where the fluxes $\bar{F}$ and $\bar{G}$ can be approximated by discrete fluxes in each direction, as in (5.3) or (5.4).

Upwind method: Examples of the two-dimensional case

Example

The first example is the simplest: a simplified two-dimensional elliptic problem as (3.1), a *quarter of the five spot* configuration as in definition 3 and homogeneous Neumann boundary conditions:

$$\left\{ \begin{array}{ll} \nabla \cdot u = q & \text{in } \Omega = [0, 1] \times [0, 1] \\ u \cdot n = 0 & \text{on } \partial\Omega \\ -K \nabla p = u & \text{(Darcy velocity)} \end{array} \right. .$$

With a constant absolute permeability $K(x, y) = 1$, source term $\tilde{q} = 1$, injection well at cell $(1, 1)$ and production well at (N, M) .

Upwind method: Examples of the two-dimensional case

Example

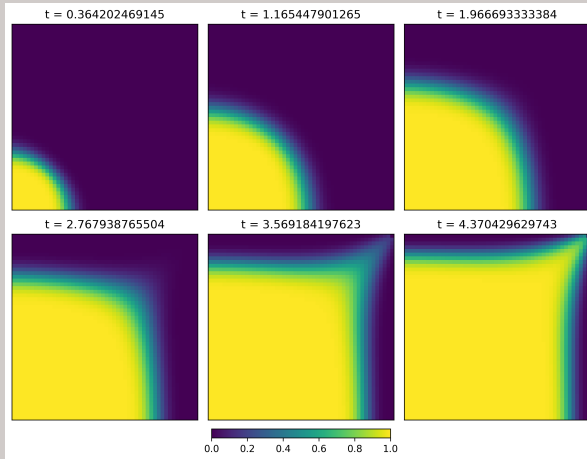
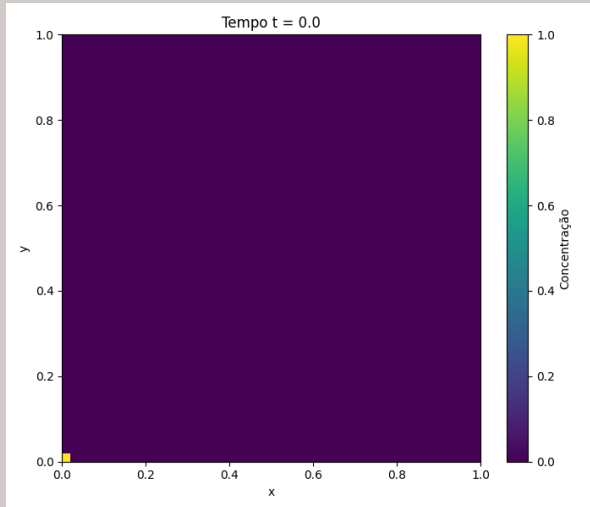


Figure: Contaminant dispersion panel for the example.

Upwind method: Examples of the two-dimensional case

Example



Upwind method: Examples of the two-dimensional case

Example

In this second example, given a simplified two-dimensional elliptic problem as (3.1), a *quarter of the five spot* configuration, permeability field and pressure and velocity fields visualized below:

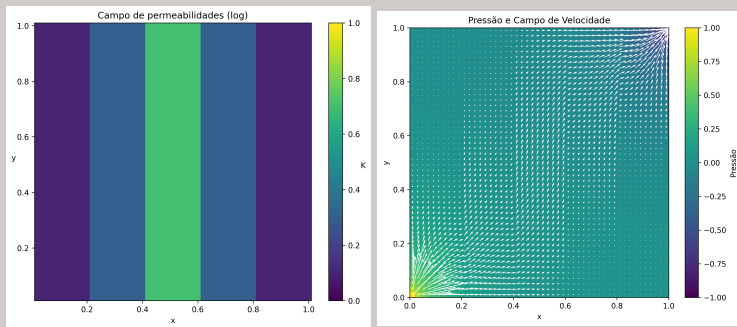


Figure: Permeability plots and pressure with non-normalized velocities.

Upwind method: Examples of the two-dimensional case

Example

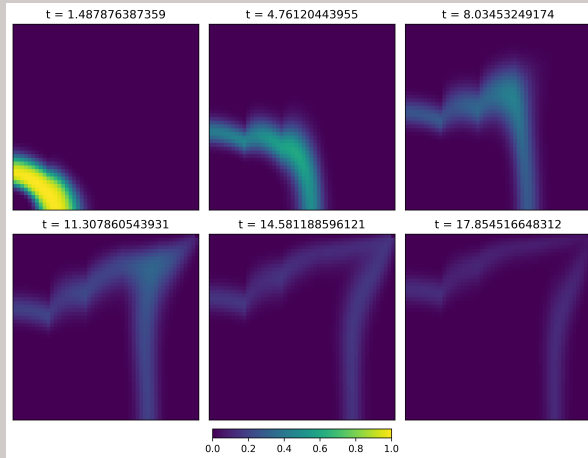
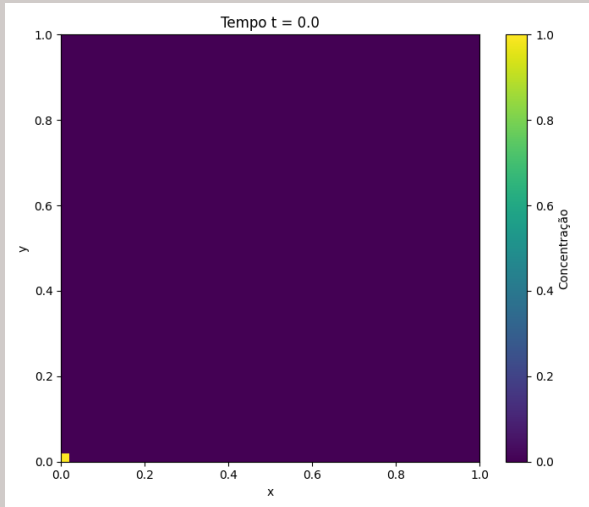


Figure: Contaminant dispersion panel for the example.

Upwind method: Examples of the two-dimensional case

Example



Presentation structure

- ① General definitions
- ② Single-phase flow modeling
- ③ Finite volume method for elliptic equations
- ④ Introduction to passive transport problems in porous media
- ⑤ Finite volume method for hyperbolic equations
- ⑥ References

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Obrigado!

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